

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

MACHINE LEARNING [ECE-4409]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss classification and regression problem used in machine learning with necessary equations and notations. (4)
 - A)
 - B) Contrast between supervised and unsupervised learning algorithms with one example each. (3)
 - C) Discuss on statistical learning theory (3)
- 2) Discuss the concept of sequential backward search with respect to feature selection? (4)
 - A)
 - B) Discuss the formulation of Support Vector Machine classification problem? (4)
 - C) Discuss on radial basis function kernel used in SVM classification with necessary equation? (2)
- 3) Explain the backpropagation algorithm used in training of artificial neural networks? (4)
 - A)
 - B) Discuss the concept of maximum likelihood estimation used in Bayesian learning (4)
 - C) Explain the decision rule based on the class conditional density for a two-class classification problem. (2)
- 4) Discuss 10-fold cross validation with a neat schematic? (4)
 - A)
 - B) Discuss on Student's t-test with computational steps? (4)
 - C) Explain precision and recall with respect to performance of a classifier? (2)

- 5) Discuss the concept of reward and policy with reference to reinforcement learning? (4)
- A)
- B) Discuss in brief the hierarchical clustering methods used in Machine Learning? (3)
- C) Discuss the histogram estimator used for density estimation in machine learning? (3)

-----End-----